

LEAD
LAG

```
SELECT *,  
LEAD(SALARY) OVER (ORDER BY SALARY DESC) as next_sal  
LAG(SALARY) OVER (ORDER BY SALARY DESC) as prev_sal  
FROM EMPLOYEES;
```

```
SELECT *,  
LEAD(SALARY) OVER (ORDER BY SALARY DESC) as next_sal,  
LAG(SALARY) OVER (ORDER BY SALARY DESC) as prev_sal  
FROM EMPLOYEES;
```

```
CREATE TABLE sales (  
sale_date DATE,  
product_id INT,  
product_name VARCHAR(100),  
units_sold INT  
);
```

```
INSERT INTO sales (sale_date, product_id, product_name, units_sold)  
VALUES
```

```
('2024-08-15', 1, 'Laptop', 4),  
('2024-08-16', 1, 'Laptop', 6),  
('2024-08-17', 1, 'Laptop', 5),  
('2024-08-18', 1, 'Laptop', 7),  
('2024-08-19', 1, 'Laptop', 4),  
('2024-08-20', 1, 'Laptop', 5),  
('2024-08-21', 1, 'Laptop', 8),  
('2024-08-22', 1, 'Laptop', 3),  
('2024-08-23', 1, 'Laptop', 7),  
('2024-08-24', 1, 'Laptop', 6),  
('2024-08-25', 1, 'Laptop', 9),  
('2024-08-26', 1, 'Laptop', 7),
```

```
('2024-08-15', 2, 'Smartphone', 15),  
('2024-08-16', 2, 'Smartphone', 18),  
('2024-08-17', 2, 'Smartphone', 12),  
('2024-08-18', 2, 'Smartphone', 20),  
('2024-08-19', 2, 'Smartphone', 17),  
('2024-08-20', 2, 'Smartphone', 14),  
('2024-08-21', 2, 'Smartphone', 19),  
('2024-08-22', 2, 'Smartphone', 13),  
('2024-08-23', 2, 'Smartphone', 21),  
('2024-08-24', 2, 'Smartphone', 16),  
('2024-08-25', 2, 'Smartphone', 18),  
('2024-08-26', 2, 'Smartphone', 20),
```

```
('2024-08-15', 3, 'Tablet', 8),  
('2024-08-16', 3, 'Tablet', 10),  
('2024-08-17', 3, 'Tablet', 7),  
('2024-08-18', 3, 'Tablet', 12),  
('2024-08-19', 3, 'Tablet', 9),  
('2024-08-20', 3, 'Tablet', 8),  
('2024-08-21', 3, 'Tablet', 11),  
('2024-08-22', 3, 'Tablet', 6),  
('2024-08-23', 3, 'Tablet', 9),  
('2024-08-24', 3, 'Tablet', 10),  
('2024-08-25', 3, 'Tablet', 13),  
('2024-08-26', 3, 'Tablet', 12);
```

```
select sale_date,  
product_id,  
product_name,  
units_sold,  
lag(units_sold,1) over (partition by product_id order by sale_date)  
as previous_day_sales,  
lead(units_sold,1) over (partition by product_id order by sale_date)  
as next_day_sales  
from sales;
```

-- lead with order by asc

-- lag with order by desc

```
select sale_date,  
product_id,  
product_name,  
units_sold,  
lag(units_sold,1) over (partition by product_id order by sale_date)  
as lag_result,  
lead(units_sold,1) over (partition by product_id order by sale_date desc)  
as lead_result  
from sales;
```

=====

```
CREATE TABLE social_media_followers (  
user_id INT,  
user_name VARCHAR(100),  
month DATE,  
linkedin_followers INT,  
twitter_followers INT,  
instagram_followers INT,  
youtube_followers INT  
);
```

```

INSERT INTO social_media_followers (user_id, user_name, month, linkedin_followers,
twitter_followers, instagram_followers, youtube_followers) VALUES
(1, 'Rajesh', '2023-08-01', 500, 1200, 800, 900),
(1, 'Rajesh', '2023-09-01', 550, 1300, 850, 950),
(1, 'Rajesh', '2023-10-01', 600, 1400, 900, 1000),
(1, 'Rajesh', '2023-11-01', 660, 1500, 950, 1050),
(1, 'Rajesh', '2023-12-01', 720, 1600, 1000, 1100),
(1, 'Rajesh', '2024-01-01', 790, 1700, 1060, 1160),
(1, 'Rajesh', '2024-02-01', 860, 1800, 1120, 1220),
(1, 'Rajesh', '2024-03-01', 940, 1900, 1190, 1290),
(1, 'Rajesh', '2024-04-01', 1020, 2000, 1260, 1360),
(1, 'Rajesh', '2024-05-01', 1100, 2100, 1330, 1430),
(1, 'Rajesh', '2024-06-01', 1190, 2200, 1410, 1510),
(1, 'Rajesh', '2024-07-01', 1280, 2300, 1490, 1590),
(1, 'Rajesh', '2024-08-01', 1380, 2400, 1580, 1680),
(2, 'Anjali', '2023-08-01', 800, 500, 600, 700),
(2, 'Anjali', '2023-09-01', 820, 510, 620, 710),
(2, 'Anjali', '2023-10-01', 840, 520, 640, 720),
(2, 'Anjali', '2023-11-01', 860, 530, 660, 730),
(2, 'Anjali', '2023-12-01', 880, 540, 680, 740),
(2, 'Anjali', '2024-01-01', 900, 550, 700, 750),
(2, 'Anjali', '2024-02-01', 920, 560, 720, 760),
(2, 'Anjali', '2024-03-01', 940, 570, 740, 770),
(2, 'Anjali', '2024-04-01', 960, 580, 760, 780),
(2, 'Anjali', '2024-05-01', 980, 590, 780, 790),
(2, 'Anjali', '2024-06-01', 1000, 600, 800, 800),
(2, 'Anjali', '2024-07-01', 1020, 610, 820, 810),
(2, 'Anjali', '2024-08-01', 1040, 620, 840, 820);

```

-- calculate % of total follower gains for each user month on month

```

select *,
lead(total_followers,1) over (partition by user_id order by month) as
next_mnth,
lag(total_followers,1) over (partition by user_id order by month) as prev_mnth
from (select user_id,
user_name,
month,
linkedin_followers + twitter_followers + instagram_followers +
youtube_followers
as total_followers
from social_media_followers) temp
with cte1 as (select user_id,
user_name,
month,
linkedin_followers + twitter_followers + instagram_followers +
youtube_followers
as total_followers

```

```
from social_media_followers),
cte2 as (select *,
lead(total_followers,1) over (partition by user_id order by month) as
next_mnth,
lag(total_followers,1) over (partition by user_id order by month) as prev_mnth
from cte1)
select *,
round((total_followers - prev_mnth)*100 / prev_mnth ,2 ) as gain_percent
from cte2;
```